# Lab 5 – Contribution Journal

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## Introduction

For Lab 5, my contribution focused on preparing the Titanic dataset for machine learning. I worked through the data preparation tasks, wrote Python code, reviewed the output, and considered why each preprocessing technique is useful. The lab helped me understand that data often needs to be cleaned and converted before a machine learning model can be used.

## My Specific Contributions

First, I handled missing values in the Age column. I calculated the median age and used the fillna() method to replace missing entries. I then checked the column with isnull().sum() to verify that no missing Age values remained. I chose the median because it is less affected by extreme values than the mean.

Next, I used pd.get_dummies() to encode the Sex and Embarked columns. I set drop_first=True and saved the result in encoded_df. This converted the categorical information into numerical indicator columns and removed the original columns that were encoded.

I also applied feature scaling to the Age and Fare columns. I imported StandardScaler, created a scaler, selected the two numerical columns, and used fit_transform() to standardize their values. Finally, I displayed the first few rows to inspect the data that was transformed. I completed the knowledge check by explaining median imputation, one-hot encoding, and why decision trees usually do not require feature scaling.

## My Learning and Reflection

This lab helped me connect data preparation methods to the machine learning workflow. I learned that imputing missing values can preserve rows that might otherwise be deleted, while one-hot encoding makes categorical information usable by models that require numerical input. I also learned that standardization puts numerical features on a comparable scale, which can be useful for some algorithms. The lab showed me that preprocessing choices depend on the dataset and the model being used, rather than every technique being necessary in every situation.

## Limitations and Future Contribution

In future projects, I would improve my contribution by checking all columns for missing values and deciding how to handle each one, rather than focusing only on Age. I would also verify the data types of the encoded columns and compare model results before and after scaling when appropriate. For a real predictive task, I would fit preprocessing steps using the training data only to avoid data leakage.

## Conclusion

Overall, my contribution to Lab 5 was to complete and review the data preparation steps for the Titanic dataset. I practiced filling missing values, encoding categorical features, scaling numerical columns, and explaining the purpose of these techniques. This work strengthened my Python data-analysis skills and helped me understand how thoughtful preprocessing supports machine learning.